

F1 Qualifying Explorer

Technical Specification · v1.0 · Australian GP 2026

1. Technology Stack

Layer	Technology	Notes
Backend	Python 3.11 · FastAPI	Async, auto OpenAPI docs
Frontend	React 18 · Vite	SPA, autocomplete UI
Data	JSON (local file)	Loaded once at startup into memory
Testing	pytest	Unit tests on core data logic
Dev server	Uvicorn	ASGI, hot-reload in dev

2. Source Data — Key Fields

Fields consumed from `session_laptimes.json`:

Field	Type	Usage
Driver	str	3-letter driver code (e.g. HUL). Used as primary lookup key.
Session	str	Q1 Q2 Q3 — qualifying session identifier.
LapTime	float null	Lap duration in seconds. Null laps are excluded.
IsAccurate	bool	False = lap flagged as unreliable; excluded from best-time calc.
PitOutTime / PitInTime	str null	Non-null = outlap or inlap; excluded from best-time calc.
Position	int null	Final classified qualifying position.
Compound	str	Tyre compound (informational).

3. Lap Validity Rules

A lap is considered **valid** for best-time selection if ALL of the following hold:

#	Condition	Field checked
1	LapTime is not null / NaN	LapTime
2	IsAccurate == True	IsAccurate
3	Not an outlap (no PitOutTime)	PitOutTime
4	Not an inlap (no PitInTime)	PitInTime
5	LapTime > 0	LapTime

Best lap per session = min(LapTime) over all valid laps for that driver & session.

4. API Endpoints

4.1 GET /api/drivers

Returns list of all driver codes present in the dataset. Used to populate autocomplete.

	Detail
Method	GET
Auth	None
Response 200	{"drivers": ["BOR", "HUL", "NOR", ...]}
Response 500	{"detail": "Data not loaded"}

4.2 GET /api/driver/{code}

Returns best valid lap per session and final qualifying position for a given driver.

	Detail
Method	GET
Path param	code — 3-letter driver code, case-insensitive, trimmed
Auth	None
Response 200	DriverResult schema (see §6)
Response 404	{"detail": "Driver 'XYZ' not found in dataset"}
Response 422	FastAPI validation error (invalid path param type)
Response 500	{"detail": "Data not loaded"}

5. Pydantic Schemas

SessionResult

Field	Type	Description
session	str	Session label: Q1, Q2, or Q3
best_lap_time	float None	Best valid lap time in seconds. None if driver had no valid laps.
best_lap_time_fmt	str None	Formatted as "M:SS.mmm". None if no valid lap.
participated	bool	True if driver has any lap (valid or not) in this session.

DriverResult

Field	Type	Description
driver_code	str	Normalised 3-letter driver code (uppercased).
qualifying_position	int None	Final classified position. None if not present in dataset.
sessions	list[SessionResult]	One entry each for Q1, Q2, Q3 (always 3 items).

6. Error Handling

Scenario	Behaviour
JSON file missing / unreadable	App fails to start with logged error; all endpoints return 500.
Malformed JSON	Caught at load time; app logs parse error and returns 500 on requests.
Null / missing lap fields	Lap is skipped silently during validation (does not crash).
Unknown driver code	404 with descriptive message.
Driver in dataset but no valid laps in a session	participated=False or best_lap_time=None for that session.
User input — wrong case / whitespace	Normalised server-side (strip + upper) before lookup.

7. Test Coverage (pytest)

Test	What it covers
test_valid_lap_filter	Laps with IsAccurate=False, pit times, null LapTime are excluded.
test_best_lap_selection	Min valid lap time selected correctly per session.
test_unknown_driver_returns_none	Missing driver code returns None result, not an exception.
test_no_valid_laps_session	Session with only invalid laps returns best_lap_time=None.
test_qualifying_position	Position field correctly extracted from dataset.
test_lap_time_formatting	Seconds correctly formatted to M:SS.mmm string.
test_malformed_data_graceful	Rows missing expected fields do not raise exceptions.

8. Frontend Components

Component	Props / State	Responsibility
App.jsx	— (root)	State: selectedDriver, driverList. Fetches /api/drivers on mount.
DriverSearch.jsx	drivers[], onSelect(code)	Controlled input with filtered dropdown autocomplete.
ResultsCard.jsx	result: DriverResult	Displays position + Q1/Q2/Q3 best laps in a clean card layout.

API base URL configured via VITE_API_URL env var (default: http://localhost:8000).